

packages/graph-view/scripts/benchmark-layout.mjs

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Overview

A new dev script `packages/graph-view/scripts/benchmark-layout.mjs` (lines 1-80) has been added to measure ELK layout performance for synthetic Stage 1 graphs.

Key changes

- **Imports:** `performance` from `node:perf_hooks` (R16) and `ELK` from `elkjs/lib/elk.bundled.js` (R17).
- **Constants:** `DEFAULT_NODE_WIDTH 280` and `DEFAULT_NODE_HEIGHT 120` (R19-R21), matching production defaults in `src/utils/layout.ts`.
- **Utility:** `fillDims` (R30-R37) copies nodes and assigns default dimensions if missing, mirroring the production repair step.
- **Layout wrapper:** `applyElkLayout` (R40-R43) repairs the input with `fillDims` and calls `elk.layout`.
- **Synthetic graph generator:** `makeGraph` (R51-R66) builds a graph of container nodes and sparse edges, modeling Stage 1 shape.
- **Benchmark routine:** `bench` (R68-R75) times a single layout run and logs elapsed ms.
- **Execution:** runs `bench` for 500, 1000, and 3000 nodes (R78-R80). Comments (R9-R12) state performance targets: <200 ms at 500 nodes, <500 ms at 3000 nodes.

Impact

- No production code changes; the file is a standalone dev script.
- Requires `elkjs` and Node's `perf_hooks`; ensure `elkjs` is in `devDependencies`.
- Uses top-level `await`; requires Node 14+ with ES-module support.

Risks & follow-ups

- Verify `elkjs/lib/elk.bundled.js` remains a valid import path; update if the library changes.
- Confirm CI environments support top-level `await` and `perf_hooks`.
- Ensure the script is not invoked in production or CI pipelines, as it can be time-consuming.